Shyamsundhar Yathirajam

Data Scientist | Machine Learning Engineer

California| 760-639-8222 | shyamsundhar@myemailjob.com |LinkedIn |Git Hub

SUMMARY

Experienced Machine Learning and Generative AI Engineer with 4+ years of expertise in Data Modeling, Wrangling, Mining, Statistical and Mathematical Modeling, and Computer Vision. Skilled in building predictive models (Regression, Random Forests, SVM, etc.) and optimizing deep learning architectures (CNNs, LSTMs, transformers) using PyTorch and TensorFlow. Proficient in Generative AI, LLMs, quantization techniques for efficient on-device inference, and scalable language translation systems. Hands-on experience with GPU-based acceleration and embedded systems, including NVIDIA Jetson platforms, enabling real-time inference and edge deployments. Adept in Python (NumPy, Pandas, SciPy), AWS services (EC2, S3, EMR), and data visualization tools like Tableau and R Shiny. Recognized for innovative AI research, focusing on model compression, efficient inference, and hardware-aware optimization.

SKILLS

Methodologies: SDLC, Agile, Waterfall
Language: Python, R, SQL, SAS, C#, Scala, C/C++, Java, Javascript
IDEs: Visual Studio Code, PyCharm, Jupyter Notebook
Statistical Methods: Hypothetical Testing, ANOVA, Time Series
Machine Learning: Regression analysis, Bayesian Method, Decision Tree, Random Forests, Support Vector Machine, Neural Network, Sentiment Analysis, K-Means, KNN, Classification, SVM, Naive Bayes, NLP, LLM, LVM, CNN, XGBoost, Julia, Deep Learning, Predictive Models, LangChain, LlamaIndex, Haystack, Pinecone, Cassandra, Qdrant, TensorRT, MLFlow, Vertex AI, Kubeflow
Packages: NumPy, Pandas, Matplotlib, SciPy, ggplot2, Scikit-Learn, PyTorch (CUDA), TensorFlow, Keras, Spark
Visualization Tools: Tableau, Power BI, Microsoft Excel
Cloud Technologies: AWS, GCP (including Google BigQuery), Docker, Kubernetes, Azure, NVIDIA Jetson platforms.
Database: MySQL, SQL Server, Oracle, MongoDB, SAP ECC, S/4 HANA Migration
Software/Other Skills: Jira, Data Cleaning, Data Wrangling, Critical Thinking, Communication Skills, Presentation Skills, Problem-solving, Decision-Making, EDA, Communication Skills, Databricks, Data Visualization, Predictive Analytics, Pattern Recognition, JMP, Data Integrity, Quantitative Data, Data Science, Statistics, Statistical Analysis, Data Analytics, Data Modeling, Data Reporting, Snowflake, Data Extraction
Generative AI Tools: Hugging Face Transformers, OpenAI API, LangChain, GPT Fine-Tuning

WORK EXPERIENCE

Ally Financial, USA | Data Scientist — Jan 2024 - Current

- Designed and deployed a cutting-edge e-commerce website leveraging Large Language Models (LLMs) to enhance product search and recommendation systems, improving user experience and engagement.
- Developed a hybrid recommendation system integrating collaborative and content-based filtering, achieving a 15% boost in conversion rates and 20% improvement in product discovery through advanced techniques like matrix factorization (ALS, SVD).
- Created elegant and customizable visualizations using ggplot2 in R, adhering to the principles of the grammar of graphics.
- Developed and fine-tuned generative AI models, leveraging Large Language Models (LLMs) for text generation, summarization, and conversational AI applications.
- Utilized machine learning algorithms, including linear regression, multivariate regression, Naive Bayes, Random Forests, K-means, and KNN, for predictive modeling and data-driven insights.
- Utilized advanced model compression and quantization techniques to optimize generative AI models for efficient on-device.
- Leveraged AWS services (S3, DynamoDB, Lambda, EC2, and SageMaker) to deploy, tune, and monitor machine learning models, ensuring scalable and efficient cloud-based operations.
- Implemented sentiment analysis pipelines using spaCy and NLTK, achieving 90% accuracy in classifying customer feedback
- Developed containerized ML solutions using Docker and deployed them on Kubernetes clusters for scalable and reliable operations.
- Created real-time performance reports in Tableau and customized SQL queries in MS SQL Management Studio for actionable insights.
- Optimized data processing with Apache Spark and Hadoop, integrating Hive workflows to reduce processing time by 20%.
- Performed data cleaning, and analysis using Spark's MLLib, leveraging deep learning frameworks for improved model performance.
- Ensured seamless machine learning model deployment and updates with CI/CD pipelines, enhancing operational efficiency.

California State University, USA | Research Data Scientist — June 2023 - Dec 2023

- Enhanced data processing times by 55x using hyperdimensional computing models on both CPU and GPU (CUDA).
- Achieved 98% performance improvement of CNN algorithms by researching existing methods, coding new algorithms in Python.
- Applied Fourier transforms and signal processing methods to analyze wind farm oscillation data, improving feature extraction from complex datasets.
- Designed a chatbot for healthcare providers using Rasa, enabling automated symptom triage and reducing initial response times by 50% in a project.
- Created customized ML pipelines for feature engineering, model tuning, and deployment to production environments.
- Published 6 research findings in top-tier conferences like IEEE, showcasing novel approaches in image segmentation and pattern recognition.
- Performed A/B testing to compare the effectiveness of different machine learning models for data-driven insights.
- Leveraged GCP services, including Google Cloud Storage and BigQuery, for scalable data solutions and efficient processing of large datasets, while continuously monitoring and evaluating model performance.
- Designed custom hardware-in-loop optimization workflows for model deployment on edge devices, including NVIDIA Jetson platforms.

Rlogical Techsoft Pvt. Ltd, India | Data Scientist **Jan 2020 - July 2022**

- Conducted thorough testing and validation of data science models and solutions before deployment, following a structured and rigorous approach typical of Waterfall methodologies.
- Implemented automated machine learning (AutoML) techniques to optimize model selection and hyperparameters.
- Used pandas, NumPy, seaborn, SciPy, Matplotlib, scikit-learn, and NLTK in Python for developing various machine learning algorithms.
- Integrated generative AI capabilities into data-driven workflows, enabling automated content generation, personalization, and insights extraction.
- Data Manipulation and Aggregation from different sources using Nexus, Toad, Business Objects, Power BI, and Smart View.
- Implemented Classification using supervised algorithms like Logistic Regression, Decision trees, KNN, Naive Bayes.
- Built scalable generative AI-powered systems for real-time recommendations, sentiment analysis, and natural language processing (NLP) tasks.
- Evaluated the performance of Various Classification and Regression algorithms using R language to predict the future power.
- Collaborated with cross-functional teams to integrate Databricks solutions into existing data pipelines, ensuring seamless data flow.
- Implemented supervised and unsupervised learning algorithms, including LLM (Latent Dirichlet Allocation), to uncover patterns in large datasets.
- Developed a deep learning model for the industrial decision-making process through Keras - Models such as RNN and LSTM were also implemented and deployed on a cloud environment like GCP for continuous monitoring of model performance.
- Deployed and monitored generative AI models on cloud platforms using AWS (SageMaker, Lambda) and GCP (Vertex AI) for seamless scalability and reliability.
- Implemented backup and recovery strategies for SQL Server databases, ensuring data availability and integrity in case of system failures or disasters.

EDUCATIONS

PhD in Computer Science | Stanford University (Application Submitted and received a positive response from an Advisor)

Master's in computer science | California State University San Marcos, San Marcos, CA

Bachelor's in computer science | Jawaharlal Nehru Technological University Hyderabad, Hyderabad, Telangana

Diploma in computer science | Osmania University, Hyderabad, Telangana

PUBLICATIONS

- Yathirajam Shyam, Peighambari Arash, Ruben Roberts, Sreedevi Gutta, Hamed Nademi, Morris Justin, and Ali Ahmadinia, "Windfarm Forced Oscillation Detection Using Hyper dimensional Computing" IEEE International Conference on Big Data (BigData), 2023.
- Yathirajam Shyam Sundhar, Sreedevi Gutta, "Efficient glioma grade prediction using learned features extracted from convolutional neural networks" in Journal of Medical Artificial Intelligence. 2024.
- Yathirajam Shyam Sundhar, Sreedevi Gutta, "Improved Glioma Grade Prediction using Mean Image Transformation" in AIME 2024: 22nd International Conference on Artificial Intelligence in Medicine, 2024.
- Yathirajam Shyam Sundhar, Sreedevi Gutta, "Glioma Grade Prediction using Hyper-dimensional Computing" in AIME 2024 Conference.
- "Advancing Multilingual Language Translation with Large Language Models and Reinforcement Learning", IEEE GreenTech 2024.
- Yathirajam Shyam Sundhar, Peighambari Arash, Nikil Balaji, Owen Man, Sreedevi Gutta, Hamed Nademi, Morris Justin, and Ali Ahmadinia, "Improved Subsynchronous Frequency Oscillations detection in Wind Farms using AI-based Fourier Transformation and Advancede Metrics" in IEEE GreenTech 2025 Conference

AWARDS

- Nominated for **"Dean's Awards 2024"** at **CSUSM** for innovative research and publications in the field of AI in different domains.
- Awarded with **"Student Faculty Collaboration Award 2024"** at **CSUSM** for the research in Hyper dimensional computing and AI.
- Awarded with "**State Level Best Dancer**" in a state level Dance Competition in 2018.

PROJECTS

- **AI-Powered Customer Support Chatbot**: Developed a scalable AI chatbot using LLMs for real-time customer support, enhancing resolution rates and integrating with CRM systems for personalized responses.
- **Multi-Agent Decision-Making System**: Designed and implemented an agentic AI framework using LangChain and Graph-based reasoning (Gephi) to enable dynamic decision-making and collaboration across multi-agent systems, optimizing task allocation and resource utilization in complex environments.
- **Predictive Maintenance System for Manufacturing**: Designed a time-series-based predictive maintenance system using LSTMs and BigQuery, reducing downtime by 20% and optimizing health tracking equipment.